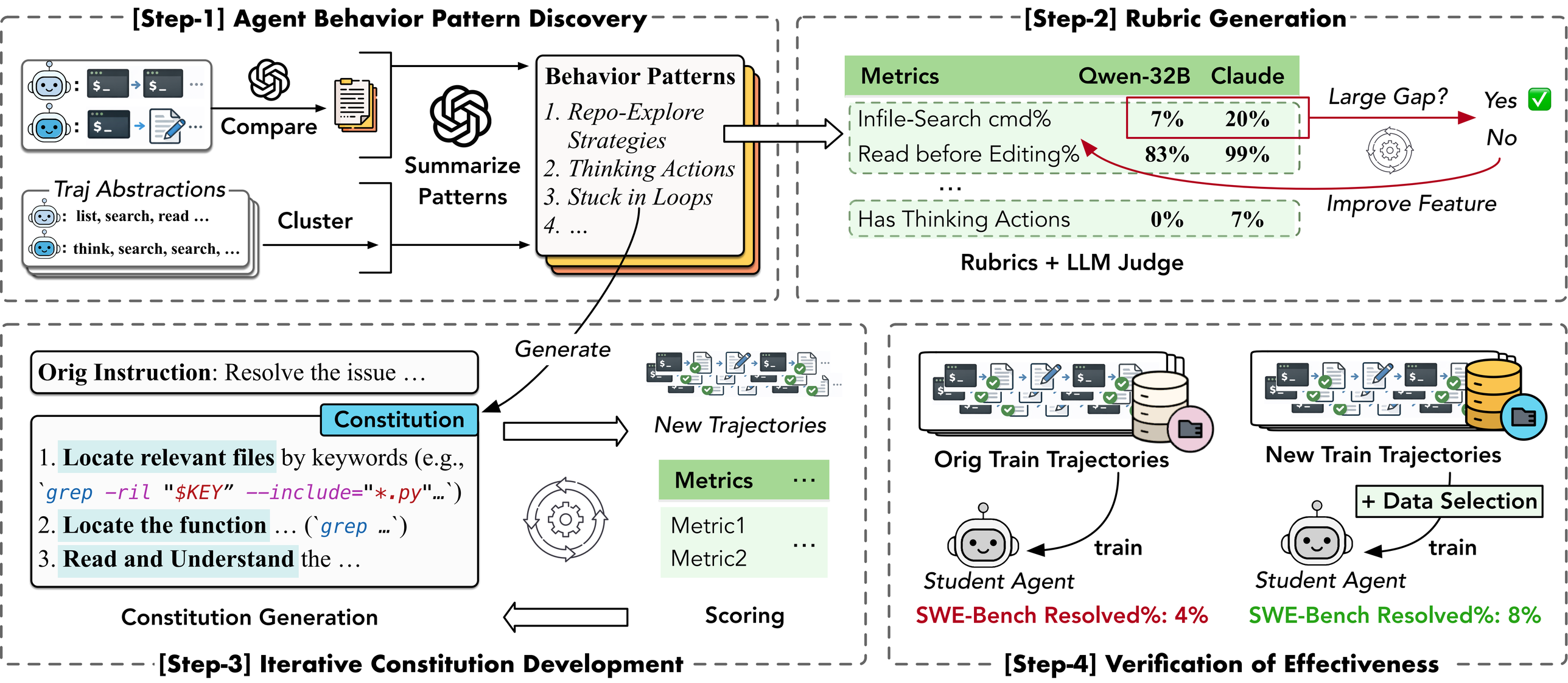
[Step-1] Agent Behavior Pattern Discovery
Compare
Traj Abstractions
list, search, read …
think, search, search, …
Cluster
Summarize Patterns
Behavior Patterns
1. Repo-Explore Strategies
2. Thinking Actions
3. Stuck in Loops
4. …
[Step-2] Rubric Generation
Metrics
Qwen-32B
Claude
Infile-Search cmd%
7%
20%
Read before Editing%
83%
99%
…
Has Thinking Actions
0%
7%
Large Gap?
Yes
No
Improve Feature
Rubrics + LLM Judge
Orig Instruction: Resolve the issue …
Generate
Constitution
1. Locate relevant files by keywords (e.g., `grep -ril "$KEY" --include="*.py"…`)
2. Locate the function … (`grep …`)
3. Read and Understand the …
Constitution Generation
New Trajectories
Metrics
Metric1
Metric2
Scoring
[Step-3] Iterative Constitution Development
Orig Train Trajectories
train
Student Agent
SWE-Bench Resolved%: 4%
New Train Trajectories
+ Data Selection
train
Student Agent
SWE-Bench Resolved%: 8%
[Step-4] Verification of Effectiveness